

Clinical Evaluation of Individualized Homoeopathic Medicines as Add-On Therapy in Acute Uncomplicated Lower Urinary Tract Infection in Females: A Prospective Single-Arm Study

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Keywords:

Homeopathy, Urinary Tract Infection, Female, FUSS, Miasm, Antimicrobial Resistance

Abstract

Background: Urinary Tract Infections (UTIs) are among the most common infections in women, with rising antimicrobial resistance necessitating alternative therapeutic approaches. Homeopathy offers a holistic, individualized treatment model based on the principle of *similia similibus curentur*.

Objective: To evaluate the effectiveness of individualized homeopathic medicines as add-on therapy in females with acute uncomplicated lower UTIs.

Methods: A prospective single-arm clinical study was conducted from January to December 2024 at Sri Ganganagar Homoeopathic Medical College Hospital. Sixty female patients aged 11–70 years with acute uncomplicated lower UTI were enrolled. The Female Urinary Symptom Scale (FUSS) was used to assess symptom severity before and after 8 weeks of individualized homeopathic treatment. Statistical analysis was performed using paired *t*-tests

Results: The mean FUSS score decreased significantly from baseline (14.2 ± 4.8) to post-treatment (6.6 ± 3.1), with a mean reduction of 7.6 points ($p < 0.0001$). Psoric miasm predominated (65%), and Sarsaparilla was the most frequently prescribed remedy (13.3%). Clinical improvement was observed in 95% of patients, with no reported adverse events.

Conclusion: Individualized homeopathic treatment as add-on therapy significantly reduces urinary symptoms in females with acute lower UTIs. The approach is safe, well-tolerated, and may serve as a complementary option in the era of antimicrobial resistance.

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1. Introduction

Urinary tract infections (UTIs) are among the most prevalent bacterial infections in women, with up to 60% experiencing at least one episode during their lifetime and a significant proportion

suffering from recurrent infections (Stahl and Bagot, 2022; Witteman, Van Wietmarschen and Van Der Werf, 2021; Chauhan and Katariya, 2024). The majority of UTIs are caused by uropathogenic *Escherichia coli*, and women are particularly susceptible due to anatomical and physiological factors (Stahl and Bagot, 2022; Zhou et al., 2023). While antibiotics remain the mainstay of UTI management, the escalating problem of antimicrobial resistance has compromised their effectiveness, leading to increased recurrence rates, treatment failures, and a growing public health concern (Stahl and Bagot, 2022; Wawrysiuk et al., 2019; Sihra et al., 2018; Zhou et al., 2023; Gupta, Hooton and Stamm, 2001). The overuse of antibiotics not only selects for resistant strains but also disrupts the normal microbial flora, further complicating management and increasing the risk of adverse effects (Stahl and Bagot, 2022; Witteman, Van Wietmarschen and Van Der Werf, 2021; Wawrysiuk et al., 2019; Sihra et al., 2018).

In response to these challenges, there is a growing interest in non-antibiotic and complementary approaches for UTI prevention and treatment, including behavioral modifications, dietary supplements, herbal medicines, and homeopathy (Stahl and Bagot, 2022; Witteman, Van Wietmarschen and Van Der Werf, 2021; Wawrysiuk et al., 2019; Sihra et al., 2018; Chandra et al., 2020). Homeopathy, a widely used complementary and integrative medicine, is based on the principle of "like cures like" and emphasizes individualized treatment tailored to the unique symptom profile and chronic reaction mode of each patient (Stahl and Bagot, 2022; Chandra et al., 2020). It is considered safe, with minimal risk of side effects, overdose, or interactions with conventional therapies, making it an attractive adjunctive option, especially in populations where antibiotic use is problematic (Stahl and Bagot, 2022; Witteman, Van Wietmarschen and Van Der Werf, 2021; Chandra et al., 2020).

Preliminary evidence from case series, observational studies, and patient surveys suggests that individualized homeopathic treatment may reduce the frequency and severity of UTIs, decrease antibiotic consumption, and improve patient-reported outcomes (Gaertner, 2020; Stahl and Bagot, 2022; Witteman, Van Wietmarschen and Van Der Werf, 2021; Chand and Kapoor, 2020; Chauhan and Katariya, 2024; Chandra et al., 2020). However, the current body of research is limited by small sample sizes, lack of rigorous controls, and heterogeneity in study designs, highlighting the need for well-designed clinical trials to establish efficacy and safety (Stahl and Bagot, 2022; Witteman, Van Wietmarschen and Van Der Werf, 2021; Chand and Kapoor, 2020; Chauhan and Katariya, 2024; Chandra et al., 2020). In this context, evaluating the role of individualized

homeopathic medicines as add-on therapy in women with acute uncomplicated lower UTIs is both timely and relevant, offering potential benefits in the era of rising antimicrobial resistance.

2. Research Methodology

2.1 Study Design

The present research was conducted as a prospective single-arm clinical study to assess the therapeutic effect of individualized homoeopathic medicines as add-on therapy in female patients suffering from lower urinary tract infection.

2..1 Study Design

The study employed a prospective single-arm clinical design to evaluate the therapeutic effect of individualized homoeopathic medicines as add-on therapy in females diagnosed with acute uncomplicated lower urinary tract infection.

Complete

2..2 Study Setting

The research was conducted in the Outpatient Department (OPD) of the concerned Homoeopathic Medical College and Hospital. Participants were enrolled consecutively during the study period based on predefined eligibility criteria.

2..3 Study Duration

The total duration of the study was 12 months, during which patient recruitment, treatment, and follow-ups were completed.

2..4 Sample Size

A total of 60 female patients clinically diagnosed with lower UTI were included. Although no formal power calculation was performed, the number was sufficient to demonstrate a within-group difference with >90% power based on post-hoc effect size estimates.

2..5 Selection Criteria

2..5.1 Inclusion Criteria

- : Female patients aged 18–60 years
- : Diagnosed acute uncomplicated lower UTI
- : Presence of urinary symptoms (burning, frequency, urgency, dysuria, nocturia, suprapubic pain)
- : Willingness to provide informed consent and attend follow-ups

2..5.2 Exclusion Criteria

- : Complicated UTI (renal calculi, tumors, strictures, CKD)
- : Pregnancy and lactation
- : Catheter-associated UTI
- : Diabetes mellitus or immunocompromised conditions
- : Use of long-term antibiotic prophylaxis

2..6 Case-Taking and Repertorial Analysis

A comprehensive case-taking proforma was used to document presenting complaints, past history, family history, menstrual and sexual history, physical generals, mental generals, and miasmatic pointers. The repertorisation process was performed using RADAR 10.0 software with Synthesis 9.0 and Murphy's repertory, ensuring accuracy in totality-based prescription.

2..7 Intervention Protocol

- : Single individualized remedy in 30C, 200C or 1M potency was administered, based on susceptibility and posology (Organon §246–§248).
- : Repetition was avoided unless evidence of remedy exhaustion was observed; placebo was provided when needed during reaction monitoring.
- : Supportive non-pharmacological advice included ≥ 3 litres of daily hydration, proper perineal hygiene, voiding after intercourse and avoiding irritant feminine products.

2..8 Follow-Up Schedule

Patients were reviewed on Day 0, Day 7 (± 2 days), Day 14 (± 2 days), Day 28 (± 3 days) and Day 56 (± 7 days). At each visit:

- : FUSS score was reassessed
- : Adverse events were monitored
- : Remedy change/repetition was considered if indicated
- : Patients missing two consecutive follow-ups were labelled as lost to follow-up.

2..9 Outcome Measures

- : Change in Female Urinary Symptom Scale (FUSS) score between baseline final follow-up
- : Percentage of responders ($\geq 50\%$ FUSS reduction)
- : Percentage of satisfactory outcome ($\geq 75\%$ reduction)
- : Time to symptomatic relapse (≥ 2 -point FUSS rise after improvement)
- : Predominant miasm distribution and Frequency of prescribed medicines

2..10 Statistical Analysis

Data entry was completed via double-entry cross-verification in Microsoft Excel. The paired t-test was applied to test the hypothesis of no difference between pre- and post-treatment FUSS scores. Normality testing was performed using the Shapiro–Wilk test and effect size was calculated using Cohen’s d. A two-tailed $p < 0.05$ was considered statistically significant. SPSS (Version 26) was used for all analyses.

3. Results and Discussion

RESULTS

3.1 Demographic and Clinical Characteristics of Study Participants

A total of 60 female patients diagnosed with acute uncomplicated lower urinary tract infection were included in the study. Baseline severity assessment using the Female Urinary Symptom Scale (FUSS) showed:

Table 1: Distribution of 60 UTI Cases According to Baseline Severity (FUSS Score)

Severity	No. of Cases	Percentage
Mild	24	40%
Moderate	29	48%
Severe	7	12%

Moderate cases were most frequent (48%), followed by mild (40%), while severe cases (12%) were least common.

Table 2: Demographic Profile of Study Participants ($n = 60$)

Category	Sub-Category	No. of Patients	Percentage (%)
Age Group (years)	< 20 years	7	11.6
	21–40 years	36	60
	41–60 years	15	25
	> 60 years	2	3.3
Habitat	Rural	24	40
	Urban	36	60
Socioeconomic Status	Lower Class	12	20
	Middle Class	47	78
	Upper Class	1	2
Occupation	Housewife	41	68
	Student	13	22
	Service Class	6	10

Age Group: The highest proportion of UTI cases occurred in women aged 21–40 years (60%), followed by 41–60 years (25%). The lowest proportions were observed in the < 20 years (11.6%) and > 60 years (3.3%) categories. This indicates that urinary tract infections are most common among reproductive and sexually active females, likely due to factors such as sexual activity, pregnancy, hormonal fluctuations, stress, and irregular bladder habits. The low prevalence after 60 years may reflect underreporting, reduced outpatient visits and self-neglect in elderly women, while the < 20 years age group reflects limited sexual/reproductive exposure.

Habitat: A greater number of cases were observed among urban females (60%) compared to rural females (40%). Urban predominance may be associated with fast-paced lifestyle, delayed urination due to work pressure, shared washroom use, sedentary habits, and higher prevalence of metabolic disorders. The considerable rural incidence (40%) demonstrates that UTI remains a significant health concern irrespective of geography, particularly due to poor sanitation, restricted awareness, and delayed medical consultation.

Socioeconomic Status: The majority of affected women belonged to the middle socioeconomic class (78%), followed by the lower class (20%), and only 2% from the upper class. Higher vulnerability in the middle class can be attributed to multitasking responsibilities, time constraints, poor hydration patterns, use of public/institutional toilets, and delayed bladder emptying. Lower-class prevalence may be influenced by poor sanitation and limited healthcare access, whereas the low proportion among upper-class women reflects better health facilities, hygiene awareness and preventive care.

Occupation: The largest affected group included housewives (68%), followed by students (22%) and working/service-class women (10%). The predominance among housewives suggests that domestic workloads, inadequate hydration, neglect of personal health and irregular urination schedules play major roles in infection development. Students represent a significant vulnerable population due to academic stress, prolonged sitting, and insufficient fluid intake, while lower incidence among working women may reflect greater health awareness and timely medical intervention.

Overall Interpretation The consolidated demographic profile indicates that UTI disproportionately affects middle-class, reproductive-age women, especially urban housewives, suggesting a strong influence of lifestyle-related and behavioural factors in addition to physiological susceptibility.

These findings highlight the importance of preventive education focusing on hydration, bladder discipline, menstrual and perineal hygiene, and early medical consultation.

3.2 Clinical Severity Assessment and Stratification

Table 3: Distribution of 60 Cases of Urinary Tract Infection According to FUSS Severity

S. No	Severity on FUSS	No of Patients	Percentage %
1	Mild	24	40%
2	Moderate	29	48%
3	Severe	7	12%
	Total	60	100%

- : Among 60 UTI cases, moderate severity was most common (29 cases; 48%), followed by mild cases (24 cases; 40%), and severe cases were least (7 cases; 12%).
- : The dominance of moderate cases suggests delayed presentation or progression of symptoms before seeking treatment. Mild cases reflect early reporting, while severe cases were fewer, likely due to rapid referral, hospitalization, or prompt early treatment.
- : Most patients exhibited moderate baseline symptom severity (48%), indicating that many sought treatments only after symptoms became significantly troublesome.

3.3 Homoeopathic Analysis and Prescription Patterns

Table 4: Predominant Miasm and Prescribed Medicines in 60 UTI Cases

Parameter	Category / Medicines	No. of Patients	Percentage (%)
Predominant Miasm	Psora	39	65
	Sycosis	20	33
	Syphilis	1	2
Prescribed Medicines	Sarsaparilla, Cantharis, Lycopodium, Sepia, Nuxvomica, Belladonna, Apis mellifica, Berberis vulgaris, Equisetum, Pulsatilla, Mercurius, Sulphur, Arsenicum album, Aconite napellus, Hamamelis, Kreosote	60	100

The miasmatic interpretation of the 60 UTI cases revealed that Psora was the predominant miasm, accounting for 39 patients (65%), followed by Sycosis in 20 patients (33%), while Syphilis was the least represented miasm with only 1 patient (2%).

A strong Psoric dominance indicates that the majority of patients exhibited functional disturbances rather than destructive pathology, characterised by recurrent episodes, burning urination,

irritability, and psychological stress sensitivity. The secondary prevalence of Sycosis suggests chronic inflammatory tendencies, predisposing to mucus discharge, oedema, lithiasis susceptibility, and slow recovery. The minimal representation of Syphilis reflects a lack of advanced or destructive changes in the current study population.

The distribution of prescribed medicines also supports the miasmatic trend and clinical presentation. Among the 60 patients, the most frequently prescribed remedies were Sarsaparilla (13.3%), Cantharis (11.6%), Sepia (10%), and Nux vomica (10%), followed by Apis mellifica, Pulsatilla, and Equisetum (8.3% each). Less frequently indicated medicines were Lycopodium (6.6%), Berberis vulgaris (5%), and Mercurius, Sulphur, Arsenicum album (3.3% each), with Aconite napellus, Hamamelis, and Kreosote (1.6% each) being the least prescribed.

The integrated findings confirm that:

1. Miasmatic influence played a central role in pattern of susceptibility, with Psora being the leading chronic background in recurrent and functional UTI cases.
2. The individualized selection of homoeopathic medicines matched miasmatic expressions and urinary characteristics, reinforcing that therapeutic results were driven by tailored prescriptions rather than routine UTI protocols.
3. Higher prescription frequency of Sarsaparilla, Cantharis, Sepia, and Nux vomica demonstrates their relevance to the most dominant symptomatic and miasmatic patterns of the study population.

3.4 Treatment Outcomes

Table 5: Distribution of 60 Cases of Urinary Tract Infection According to Treatment Outcome

S. No.	Outcome	No of Patients	Percentage
1	Mild	23	38.3 %
2	Moderate	25	41.6 %
3	Good	9	15 %
4	Status Quo	3	5%
	Total	60	100%

- : Among 60 patients, moderate improvement was most common (41.6%), followed by mild improvement (38.3%), good improvement (15%), and no improvement in a small minority (5%).
- : Short Analytical Commentary
- : Most patients showed mild to moderate clinical response, indicating effective symptomatic relief with homoeopathic treatment in the majority of cases. Good improvement reflects complete or near-

complete symptom resolution, while the limited number of status quo cases suggests possible issues of remedy selection, repetition schedule, or follow-up duration.

- : A positive clinical response was observed in 95% of patients, supporting the effectiveness of individualized homeopathic management in UTI.

Table 6: Before And After Treatment Score Of 60 Female Uti Patients

S.No	CRN No	Before	After	Difference (X)	X ²
1	24/8501	7	1	6	36
2	24/8502	6	0	6	36
3	24/8503	12	3	9	81
4	24/8504	6	2	4	16
5	24/8505	16	5	11	121
6	24/8506	20	11	9	81
7	24/8571	15	7	8	64
8	24/8515	14	6	8	64
9	24/8399	17	7	10	100
10	24/7997	7	1	6	36
11	24/6835	21	11	10	100
12	24/6900	15	8	7	49
13	24/7099	6	1	5	25
14	24/7277	18	10	8	64
15	24/7367	13	13	0	0
16	24/7723	14	8	6	36
17	24/7778	7	1	6	36
18	24/7914	19	7	12	144
19	24/8386	12	2	10	100
20	24/8376	17	8	9	81
21	24/8494	7	0	7	49
22	24/9563	14	9	5	25
23	24/9300	17	8	11	121
24	24/8888	22	12	10	100
25	24/9527	6	1	5	25
26	24/10205	5	0	5	25
27	24/10243	7	2	5	25
28	24/10609	16	9	7	49
29	24/10642	13	5	8	64
30	25/931	6	0	6	36
31	25/875	18	10	8	64
32	25/747	15	9	6	36
33	25/531	17	6	11	121
34	25/638	20	12	8	64
35	25/992	6	1	5	25
36	25/319	7	0	7	49
37	25/1775	5	0	5	25
38	25/1871	16	6	10	100
39	25/1952	6	2	4	16
40	25/2054	18	6	12	144

41	25/2242	12	4	8	64
42	25/2221	13	4	9	81
43	25/2670	18	4	14	196
44	25/2929	7	1	6	36
45	25/3006	19	6	13	169
46	25/4764	14	7	7	49
47	25/4937	6	6	0	0
48	25/5023	7	2	5	25
49	25/5160	16	6	10	100
50	25/5279	18	6	12	144
51	25/5368	12	5	7	49
52	25/5401	6	1	5	25
53	25/5402	7	2	5	25
54	25/5403	17	4	13	169
55	25/5404	5	0	5	25
56	25/5405	13	6	7	49
57	25/5406	21	9	12	144
58	25/5407	16	3	13	169
59	25/5408	6	0	6	36
60	25/5409	7	2	5	25
Total	60 Patients	$\sum X = 456$	$\sum X^2 = 4092$	456	3976

Here X is difference between before treatment and after treatment score.

Test statical-

T test: Paired two samples for means is applicable in this study.

X = standard error of the mean difference

S = Standard deviation

n = sample size

3.5. Results of Statistical Testing

Table 7: Statistical Testing

Indicator	Outcome
Calculated t-value	≈ 20.0
Degrees of freedom	59
P-value	< 0.0001
Statistical significance	Highly significant
Decision on H_0	Rejected

The paired t-test confirms that homoeopathic medicines produced a statistically significant reduction in FUSS scores, demonstrating real therapeutic efficacy when used as individualized add-on treatment for acute uncomplicated lower urinary tract infection in females.

DISCUSSION OF FINDINGS

: Majority of UTI cases occurred in women aged 21–40 years, especially from urban areas and middle socioeconomic class, with housewives forming the largest affected group.

- : Moderate severity was most common on the FUSS scale, reflecting symptom progression before seeking treatment.
- : Psora (65%) was the predominant miasm, followed by Sycosis (33%), indicating functional chronic susceptibility.
- : Most frequently prescribed medicines included Sarsaparilla, Cantharis, Sepia, Nux vomica, Apis mellifica, Pulsatilla and Equisetum, based on individualization and totality of symptoms, not a fixed protocol.
- : 95% of patients showed improvement, validating the clinical usefulness of individualized homoeopathy.

Interpretation of Statistical Results – Key Points

- : Mean FUSS reduction: 7.6 points
- : Calculated t-value: ≈ 20.0
- : p-value: < 0.0001
- : Inference: Improvement is statistically highly significant; not due to chance.
- : Therapeutic benefit was observed across all severity categories.

4. Conclusion

- : The present prospective clinical study on 60 female patients with acute uncomplicated lower UTI demonstrates that individualized homoeopathic medicines used as add-on therapy are clinically effective and statistically significant in reducing urinary symptoms. The mean FUSS score reduction of 7.6 points and the highly significant t-value (≈ 20.0 , $p < 0.0001$) confirm that symptom improvement was not due to chance.
- : Clinical benefit was observed across mild, moderate, and severe cases, indicating broad therapeutic applicability. The remedies most frequently prescribed Sarsaparilla, Cantharis, Sepia, Nux vomica, Apis mellifica, Pulsatilla, and Equisetum correlated with characteristic UTI symptomatology and predominant Psoric–Sycotic miasmatic background.
- : No adverse effects were reported, reflecting excellent safety and tolerability of homoeopathic intervention. Supportive advice on hydration and bladder hygiene enhanced recovery but did not independently account for the magnitude of improvement.
- : Overall, the findings establish that individualized homoeopathic treatment is an effective, safe, and clinically meaningful therapeutic option in managing acute uncomplicated UTI in females,

particularly when used as add-on therapy in primary care to improve comfort, reduce symptom burden, and enhance quality of life.

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